
LECTURE NOTES

Probability Theory

AI512 — Introduction to Machine Learning

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Contents

1	Basic Concepts	3
1.0.1	Sample space	3
1.0.2	Event A	3
1.0.3	Probability Space:	3
1.0.4	Inclusion-Exclusion Principle:	4
1.0.5	Conditional Probability:	4
1.0.6	Independence:	4
1.0.7	Law of Total Probability:	4
1.0.8	Bayes' Rule:	4
2	Random Variables	5
2.0.1	Example of random variable	5
3	Expectance and Variance	6
3.0.1	Moments	7
3.1	Common rules for Expectance and Variance	7
4	Continuous Random Variables	8
4.1	Example	8
4.2	Cumulative distribution function (CDF)	8
4.3	Probability density function (PDF)	9
5	Common Distributions	10
6	Estimators And Bias	12
6.0.1	Estimator	12
6.0.2	Estimator Bias:	12
7	Inequalities	13

1 Basic Concepts

Intuitive meaning of probability: how often an event happens if you repeat a random experiment many times.

$$P(A) = \lim_{n \rightarrow \infty} \frac{n_A}{n}$$

1.0.1 Sample space

The **sample space** is all the possible outcomes.

- Example: Rolling one die $\rightarrow \Omega = 1, 2, 3, 4, 5, 6$

1.0.2 Event A

An **event** is just a subset of the sample space.

- Example: $A = \text{"rolling an even number"} = 2, 4, 6$

Events are what you actually care about measuring probabilities for.

We say powerset of Ω as 2^Ω , then $\sigma(\Omega) \subseteq 2^\Omega$ is an eventspace (collection of subsets of Ω). An eventspace must be closed under countable unions, countable intersections, and complements. That means:

- 1) $\bigcup_{i=1}^{\infty} A_i \in \sigma(\Omega)$
- 2) $\bigcap_{i=1}^{\infty} A_i \in \sigma(\Omega)$
- 3) $A^c \in \sigma(\Omega)$

For many discrete sample spaces such as the outcome of the roll of a pair of dice, it is possible to assume that $2^\Omega = \sigma(\Omega)$. However, for continuous sample spaces we need to define a σ -algebra that is smaller than 2^Ω . This is because an uncountable set with size 2^Ω is very unstable and breaks the rules of probability.

Example in a dart game, if we care about the precise hit of a dart arrow, this can become very complex. In dart we resort to only care about the actual points the dart arrow hits. So we make the space smaller.

A **probability measure** is a function $P : \sigma(\Omega) \rightarrow [0, 1]$ such that:

- 1) $P(\Omega) = 1$
- 2) $P(A) \geq 0$ for all $A \in \sigma(\Omega)$
- 3) If $A_1, A_2, \dots$ are disjoint events (i.e. $A_i \cap A_j = \emptyset$ for all $i \neq j$), then $P(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$

We actually want to restrict the events to a **well-behaved σ -algebra**

1.0.3 Probability Space:

A tuple defined as $(\Omega, \sigma(\Omega), P)$.

1.0.4 Inclusion-Exclusion Principle:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

This takes into account the fact that $P(A \cap B)$ is counted twice in $P(A) + P(B)$. A direct consequence of this is that $P(A \cup B) \leq P(A) + P(B)$ which is called the **union bound**.

1.0.5 Conditional Probability:

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

The intuitive meaning of this is the probability of event A given that event B has occurred. That is, the frequency of event A in the subset of trials where event B has occurred. In mathematical terms

$$P(A | B) = \lim_{n \rightarrow \infty} \frac{n_{A \cap B}}{n_B}$$

The definition of conditional probability can be rewritten as $P(A \cap B) = P(A | B)P(B)$. This is called the **product rule**.

1.0.6 Independence:

If the probability of event A is not affected by the occurrence of event B , these two events are said to be independent.

In terms of conditional probabilities we can describe this situation as $P(A|B) = P(A)$. Applied to the definition of conditional probability, this means that A and B are independent if and only if $P(A \cap B) = P(A)P(B)$ independent

$P(A \cap B) \neq P(A)P(B)$ dependent

1.0.7 Law of Total Probability:

If $B_1, B_2, \dots, B_n$ is a partition of Ω , i.e. $B_i \cap B_j = \emptyset$ for all $i \neq j$ and $\bigcup_{i=1}^n B_i = \Omega$, then

$$P(A) = \sum_{i=1}^n P(A | B_i)P(B_i)$$

1.0.8 Bayes' Rule:

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}$$

This can be derived from the definition of conditional probability as follows:

$$P(A \cap B) = P(A | B)P(B) = P(B | A)P(A) \Rightarrow P(A | B) = \frac{P(B | A)P(A)}{P(B)}$$

Assume law of total probability $\mathcal{H}$, i.e. $H_i \cap H_j = \emptyset$ for all $i \neq j$ and $\bigcup_{i=1}^n H_i = \mathcal{H}$. Also assume that D is the observed data. Then, Bayes' rule can be written as:

$$P(H_i | D) = \frac{P(D | H_i)P(H_i)}{P(D)} = \frac{P(D | H_i)P(H_i)}{\sum_{j=1}^n P(D | H_j)P(H_j)}$$

Here,

- $P(H_i)$ is our **prior belief** in a hypothesis H_i ,
- $P(H_i | D)$ is our **posterior belief** in H_i after observing the data D ,
- $P(D | H_i)$ is the **likelihood** of H_i , and $P(D)$ is the **evidence**.

The concepts above give a general framework for **statistical inference**: we start with a prior belief, collect new observations, and update our belief based on them. This approach has connections to how humans make decisions under uncertainty.

2 Random Variables

It is not always convenient to describe sets. We can facilitate this by defining a random variable.

A **random variable** is function $X : \Omega \rightarrow \Lambda$ that maps each elementary event $\omega \in \Omega$ to an element on its **range** $\lambda \in \Lambda$. We can define a probability measure on Λ as follows:

$$P_X(A) := P(\{\omega \in \Omega : X(\omega) \in A\})$$

where $A \subset \Lambda$. This is called the **induced probability measure** of X .

Intuition: measures the outcome as a number.

2.0.1 Example of random variable

Consider the experiment of tossing a fair coin three times and denote heads by H and tails by T . The sample space for this experiment is $\Omega = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$.

Because the coin is fair

$$P(H) = P(T) = \frac{1}{2}$$

Because the coin tosses are independent events, we have

$$P(HHH) = P(H)P(H)P(H) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$$

and likewise for all other outcomes. Let X be the number of heads in a sequence. Then X is a random variable with range $\Lambda = \{0, 1, 2, 3\}$. The induced probability measure of X is:

$$1) P_X(0) = P(\{TTT\}) = \frac{1}{8}$$

- 2) $P_X(1) = P(\{TTH, THT, HTT\}) = \frac{3}{8}$
- 3) $P_X(2) = P(\{HTH, HHT, THH\}) = \frac{3}{8}$
- 4) $P_X(3) = P(\{HHH\}) = \frac{1}{8}$

Let us give another example. Let us make up two random variables from the outcome of the roll of a fair die:

- 1) X is the outcome of the die itself i.e. $\Lambda = \{1, 2, 3, 4, 5, 6\}$.
- 2) Y is whether the outcome of the die is even or odd, i.e. $\Lambda = \{even, odd\}$.

Then consider the case of the joint event $P(X \leq 4 \cap Y = even) = P(\{2, 4\}) = \frac{2}{6}$.

Marginalization

- For the following $P(X = x, Y = y)$ where Y is binary (like even odd)
- Because of this we can drop one of them

$$P(X = x) = \sum_{y \in \{E, O\}} P(X = x \mid Y = y) P(Y = y)$$

- and so

$$P(X = x) = \frac{1}{2} P(X = x \mid Y = E) + \frac{1}{2} P(X = x \mid Y = O)$$

3 Expectance and Variance

The **expectation** (a.k.a. **expected value**) of a random variable X is defined as:

$$E[X] := \sum_{x \in \Lambda} x P_X(x)$$

Take the three coin tosses example from 2. Random Variables where the random variable X indicates the number of heads, $\mathbb{E}[X] = 1.5$ because $\mathbb{E}[X] = 0 \cdot \frac{1}{8} + 1 \cdot \frac{3}{8} + 2 \cdot \frac{3}{8} + 3 \cdot \frac{1}{8} = 1.5$.

The expectation of a random variable is also called its **mean**. The expectation of a random variable is a measure of its **central tendency**. However, it does not tell us anything about the **spread** of the random variable.

This can be measured by the **variance** of a random variable which is defined as:

$$\text{Var}[X] := \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

In the above example, $\text{Var}[X] = 0.75$ because $\text{Var}[X] = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = 0^2 \cdot \frac{1}{8} + 1^2 \cdot \frac{3}{8} + 2^2 \cdot \frac{3}{8} + 3^2 \cdot \frac{1}{8} - 1.5^2 = 0.75$

The square-root of the variance is called the **standard deviation** of a random variable. Standard deviation is commonly denoted by σ . In the above example, $\sigma(x) = \sqrt{0.75} = 0.866$.

3.0.1 Moments

Generally speaking, a **moment** of a random variable is defined as:

$$\mathbb{E}[(X - \mathbb{E}[X])^k]$$

where k is a positive integer. The first moment is the expectation itself. The second moment is the variance. The third moment is called the **skewness** of a random variable. It is a measure of the asymmetry of the distribution of a random variable. The fourth moment is called the **kurtosis** of a random variable. It is a measure of the heaviness of the tails of the distribution of a random variable.

3.1 Common rules for Expectance and Variance

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$$

$$\mathbb{E}[c] = c$$

- Discrete

$$\mathbb{E}[g(X)] = \sum_x g(x), P(X = x)$$

- Continuous

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x)f_X(x), dx$$

$$\mathbb{E}[aX] = a\mathbb{E}[X]$$

$$\mathbb{E}\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n \mathbb{E}[X_i]$$

$$\text{If } X \perp Y : \quad \mathbb{E}[XY] = \mathbb{E}[X] \cdot \mathbb{E}[Y]$$

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid Y]]$$

$$\text{Var}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

$$\text{and so } \mathbb{E}[X^2] = \mathbb{E}[X]^2 + \text{Var}[X]$$

$$\text{Var}(aX) = a^2 \text{Var}(X)$$

$$\text{Cov}[X, Y] := E[(X - E[X])(Y - E[Y])] = E[XY] - E[X]E[Y]$$

$$\text{Var}(AB) \neq \text{Var}(A) \cdot \text{Var}(B)$$

above only applies if they are **dependant**

$$\text{Var}(AB) = \mathbb{E}[A]^2 \text{Var}(B) + \mathbb{E}[B]^2 \text{Var}(A) + \text{Var}(A) \text{Var}(B)$$

$$\text{Var}\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n \text{Var}[X_i]$$

4 Continuous Random Variables

Now consider the case where $\Lambda = \mathbb{R}$. This means a random variable X maps outcomes $\omega \in \Omega$ to numbers in $\mathbb{R}$.

In this case, we have a **continuous random variable** with the induced probability measure:

$$P_X(A) = P(\{\omega \in \Omega : X(\omega) \in A\})$$

where $A \subset \Lambda = \mathbb{R}$.

This probability measure has the pre-image of A under X on its argument, i.e. $P_{X(A)} = P(X^{-1}(A))$ Where $X^{-1}(A)$ is the pre-image, aka all the inputs that, when put through the function X , give a numbers in A (the ones we care about).

A common situation for using $X : \mathbb{R} \rightarrow \mathbb{R}$ is where events are intervals (a, b) on the range of X . In this case, we can define a probability measure on $\mathbb{R}$ as follows:

$$P_X(a < x < b) := P(\{\omega \in \Omega : X(\omega) \in (a, b)\})$$

where $a, b \in \mathbb{R}$ and $a < b$.

4.1 Example

- Bus arrives uniformly at any time in $[0, 60]$ minutes
- $X(w) = w$ (random variable is just the arrival time)
- $A = [0, 15]$ minutes (we want to know the prop. that the bus arrives in between 15 and 45 minutes.)
- Pre-image is then $X^{-1}(A) = [15, 45]$ in Ω
- Probability is therefore:

$$P_X(A) = P(X^{-1}(A)) = P(w \in [15, 45]) = \frac{45-15}{60} = \frac{1}{2}$$

The σ -algebra of $\mathbb{R}$ by the intervals of the form (a, b) is called the **Borel σ -algebra**.

4.2 Cumulative distribution function (CDF)

We can also determine probability given that X is $\leq$ some number x :

$$F_X(x) := P(X \leq x) \quad \forall x \in \mathbb{R}$$

Note that this is a non-decreasing function (probabilities goes up as x increases).

4.3 Probability density function (PDF)

Often, we are interested in interpreting the behavior of a continuous distribution by investigating how densely probability is packed at each point x . This is the derivative of $F_X(x)$.

$$f_X(x) := \frac{d}{dx}F_X(x) \quad \forall x \in \mathbb{R}$$

Then, by the fundamental theorem of calculus, we have that :

$$F_X(a) = \int_{-\infty}^a f_X(x)dx$$

Hence an event can be described in terms of the probability density function as follows:

$$P(a < X < b) = F_X(b) - F_X(a) = \int_{-\infty}^b f_X(x)dx - \int_{-\infty}^a f_X(x)dx = \int_a^b f_X(x)dx$$

The probability density function of a continuous random variable is non-negative and integrates to 1:

$$\int_{-\infty}^{\infty} f_X(x)dx = 1$$

However, keep in mind that the probability density function can take values greater than 1, since its argument is not a probability but a probability density. The total densities of a **set** of elementary events make up a probability.

Consider that

$$P_X(X = x) = P_X(x < X < x) = \int_x^x f_X(u)du = 0$$

This means that

- There are infinitely many possible events in (a, b)
- Therefore, the probability that an event takes place at exactly (a, a) is 0.

The cumulative distribution function of two jointly distributed random variables X and Y is defined as:

$$F_{X,Y}(x, y) := P(X \leq x, Y \leq y) = \int_{-\infty}^x \int_{-\infty}^y f_{X,Y}(u, v)dvdu$$

where $f_{\{X,Y\}}(x,y)$ is the joint probability density function of X and Y . The probability density function of X can be obtained by taking the partial derivative of $F_{\{X,Y\}}(x,y)$ with respect to x :

$$f_{X,Y}(x,y) := \frac{\partial}{\partial x} F_{X,Y}(x,y) \quad \forall x,y \in \mathbb{R}$$

The expectation of a continuous random variable is defined as:

$$E[X] := \int_{-\infty}^{\infty} x f_X(x) dx$$

The definition of its variance and standard deviation are the same as in the discrete case.

$$\text{Var}[X] := \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

$$\sigma = \sqrt{\text{Var}(X)}$$

5 Common Distributions

- Benoulli
 - Binary output (coinflip, yes/no ect.)
 - $P_X(x) = p^x(1-p)^{1-x}$
 - this way $P_X(1) = p$ and $P_X(0) = 1-p$
 - Then $\mathbb{E}[X] = p$ and $\text{Var}[X] = p(1-p)$
- Beta
 - The Beta distribution elegantly captures both our **belief about the center** and our **confidence in that belief!**
 -

$$f_X(x) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1}$$

- where $\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}$ is the beta function $B(\alpha, \beta)^{-1}$
- Note that $\Gamma(x) = (x-1)!$
- $x^{\alpha-1}$: Controls the behavior near $x = 0$
- $(1-x)^{\beta-1}$: Controls the behavior near $x = 1$
- Together they create the characteristic Beta shape
 -
 - **Example:**
 - 1) Prior: Beta(1,1) - “No idea about coin bias”
 - 2) Flip coin 10 times: 7 heads, 3 tails
 - 3) Posterior: Beta(1+7, 1+3) = Beta(8,4)
 - 4) New belief: Coin is probably biased toward heads

- Multi-noulli
 - **Concrete Example: Rolling a die** If XX represents the outcome of rolling a 6-sided die:
 - $P_X(1)$ = “probability of rolling a 1”
 - $P_X(2)$ = “probability of rolling a 2”
 - $P_X(6)$ = “probability of rolling a 6”

For a fair die: $P_X(1) = P_X(2) = \dots = P_X(6) = \frac{1}{6}$ **For a biased die:** Maybe $P_{X(6)} = 0.5$ and others are 0.1 each
- Dirichlet
 -
$$f_X(x) = \frac{\Gamma(\sum_{i=1}^d \alpha_i)}{\prod_{i=1}^d \Gamma(\alpha_i)}$$
 - **Example: Website A/B/C Testing**
 - 1) **Prior:** Dirichlet(1, 1, 1) - “no preference between variants”
 - 2) **Observe:** Variant A: 20 clicks, B: 15 clicks, C: 10 clicks
 - 3) **Posterior:** Dirichlet(21, 16, 11) - “updated beliefs”
 - 4) repeat
 - 5) At the end use results to predict new observations
- Uniform
 - **The constant $\frac{1}{b-a}$:**
 - This is the **height** of the rectangular probability density
 - **Why this value?** Because the area must equal 1 (total probability)
 - Area = height $\times$ width = $\frac{1}{b-a}(b-a) = 1 \checkmark$
 - $\mathbb{E}[X]$ is the **midpoint** of the interval - so perfectly symmetric
 - $\text{Var}[X]$ Depends on the **width** of the interval
 - Wider interval $\rightarrow$ more spread $\rightarrow$ higher variance
 - **Intuitive Understanding “All outcomes equally likely”** - that’s the essence of uniform distribution.
 -
 - **Examples:**
 - **Fair die:** Each face has probability $\frac{1}{6}$ (discrete uniform)
 - **Random time:** Arrival uniformly distributed between 2:00 PM and 4:00 PM
 - **Random number generator:** Most computers generate uniform random numbers on $[0,1]$
- Gaussian
 - Mean: μ

- ▶ Standard deviation: σ

▶

$$f_X(x) = \mathcal{N}(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma^2}\right)^2}$$

- where

$$P(a < x < b) = \int_a^b f(x) \, dx$$

- Here $\frac{1}{\sqrt{2\pi\sigma^2}}$ is the normalization, meaning that

$$\int_{-\infty}^{\infty} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma^2}\right)^2} = \sqrt{2\pi\sigma^2}$$

- and so

$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma^2}\right)^2} = 1$$

•

- Multivariable Gaussian

▶

$$f_X(x) = \mathcal{N}(x|\mu, \Sigma) = \frac{1}{\sqrt{((2\pi)^2)\|\Sigma\|}} \cdot e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1} (x-\mu)}$$

•

- for the parameters

$$\mu = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \Sigma = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

6 Estimators And Bias

6.0.1 Estimator

A function that estimates a value. Very simple

Ex we can estimate a mean $\hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i$

6.0.2 Estimator Bias:

The **bias** of an estimator f is defined as $B(f) = E[f(S)] - \theta$, where $B(f)$ is Bayes Error

An estimator is said to be **unbiased** if $B(f) = 0$.

However, the sample variance $S^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \hat{X})^2$ is a biased estimator of the variance σ^2 of a random variable X . That is why is it common to use the **unbiased sample variance** $\frac{1}{n-1} \sum_{i=1}^n (X_i - \hat{X})^2$ instead of the sample variance S^2 . The change made in the denominator is called **Bessel's correction**.

Think of it like shooting at a target:

- Unbiased estimator:
 - ▶ Your shots are **centered** on the bullseye (true value)
 - ▶ Sometimes you're high, sometimes low

- ▶ **On average**, you hit the center
- Biased estimator:
 - ▶ Your shots are **systematically off-center**
 - ▶ Maybe always shooting a bit to the left
 - ▶ **On average**, you miss the target, but... in a predictable direction :O

7 Inequalities

Markov's inequality

$$P(X \geq \epsilon) \leq \frac{\mathbb{E}[X]}{\epsilon}$$

Therefore $P(X < \epsilon) > 1 - \frac{\mathbb{E}[X]}{\epsilon}$

• Key Properties

- ▶ **Minimal Requirements:** Only needs $X \geq 0$ - the most general assumption possible.
- ▶ **One-Sided Bound:** Only controls upper tail $P(X \geq \epsilon)$, not deviations around the mean.
- ▶ **No Sample Size Improvement:** The bound doesn't get better with more data.
- ▶ **Often Vacuous:** Can give meaningless bounds like $P \leq 5/3 > 1$.
- ▶ **Why It Matters:**
 - **Starting Point:** Foundation for deriving other inequalities
 - **Worst-Case Analysis:** When you know almost nothing about your random variable
 - **Quick Feasibility Check:** “Is this probability even theoretically possible?”
- ▶ **Intuition:** If you only know the average, you can't say much about extremes. Like knowing the average income in a city tells you little about how many millionaires live there.
- Chebyshev's inequality
 - ▶ With Chebyshev's inequality, we are able to describe the same problem as an expression dependent on n (sample size).
 - ▶ With this, probability of correctness will maybe increase as sample size increases
 - In other words, **the probability that our estimate is “correct” (i.e., close to the true mean) increases as the sample size grows.**
 - This is because of

$$P(|X - \mu| \geq \epsilon) \leq \frac{\sigma^2}{\epsilon^2}$$

by this as $n \rightarrow \infty$, $rhs \rightarrow 0$

- By this to get probability $\leq \delta$, you need $n \leq \frac{\sigma^2}{n\epsilon^2}$ samples

Why This Is Better:

- ▶ **Variance Matters:** If you know values are tightly clustered (small σ^2), deviations are less likely
- ▶ **Sample Size Scaling:** More data $\rightarrow$ better estimates (empirical risk gets closer to true risk)
- ▶ **Practical Bounds:** Often gives more meaningful bounds where Markov fails

Intuition: Knowing both the average AND the spread lets you make much better predictions. Like knowing both average height and that heights cluster tightly around the mean.

-
-
- WLLN (weak law of large numbers)
 - ▶ For i.i.d. samples where, mean μ and $\text{Var}(X) = \sigma^2$
 - ▶ Then sample mean ($\hat{y}_n$) converges to μ as sample size increases
 - ▶
- Hoeffding's inequality
 - ▶ Hoeffding's for intervals not just bounded by $[0,1]$
 - ▶

$$P\left(\frac{1}{n} \sum_{i=1}^n X_i - E\left[\frac{1}{n} \sum_{i=1}^n X_i\right] \geq \epsilon\right) \leq e^{-2n\epsilon^2 / \sum_{i=1}^n (b_i - a_i)^2}$$

•

To get probability $\leq \delta$, you need $n \geq \frac{\ln(\frac{1}{\delta})}{2\epsilon^2}$ samples

The exponential concentration means that as you collect more training data, your confidence that the empirical risk approximates the true risk grows **exponentially fast**, not just linearly. This is why we can make strong theoretical guarantees about generalization in machine learning!